



East West University

Department of Computer Science and Engineering

Course Code: CSE207

Course Title: Data Structure

Group members:

Ahmed: 2023-3-60-128

Anika Bushra Oyshee: 2023-3-60-546

Abdullah-Al-Zeehan: 2023-3-60-139

Nusrat Rahman Aurna: 2023-3-60-057

Kaisan Faraz Jilan: 2023-3-60-197

Khaled Khan Tahsin: 2022-3-60-232

Forhadul Amin: 2023-3-60-095

Tahmidul Islam: 2023-3-60-443

Pronob Chakraborty: 2023-3-60-438

Project Specification

Project name: Trello Project Management Dashboard

Objective

The primary goal of this project is to implement a lightweight, console-based project management tool inspired by Trello. The system allows users to manage multiple projects, where each project consists of various tasks. Tasks are organized using priority levels and can be added, removed, or viewed through an interactive, menu-driven command-line interface.

Key Features

- **Project Management**

- Create new projects.
- Delete existing projects.
- Load and save project data to and from a text file (projectlist.txt).

- **Task Management**

- Add new tasks to a selected project.
- Delete tasks from a project.
- View all tasks under a project.
- Assign and modify task priorities (Low, Medium, High).

- **Priority Handling**

- Tasks are categorized using an enumerated type Priority, supporting Low, Medium, and High levels.
- Tasks can be sorted or filtered based on their priority.

- **User Interface (CLI-Based)**

- Text-based menus guide the user through operations.
- Error messages, success prompts, and navigation help ensure smooth user interaction.

System Architecture

This system follows a modular design approach to improve code reusability, maintainability, and clarity. Major components include:

1. main.c

Acts as the entry point of the application. It initializes the main data structures and starts the UI loop.

2. ui.c / ui.h

Handles all command-line interface operations. Provides menus for project selection, task management, and priority setting.

Sample Code(ui.h):

```
1  #ifndef UI_H
2  #define UI_H
3
4  void start_interface();
5  void display_main_menu();
6  void display_taskList_menu();
7  void display_task_menu();
8
9  void clear_screen();
10 void pause_and_clear();
11
12 #endif
```

3. project.c / project.h

Implements data structures and functions for creating, deleting, loading, and saving projects. Each project maintains its own task list.

Sample Code(project.h):

```
1  #ifndef PROJECT_H
2  #define PROJECT_H
3
4  #include <stdint.h>
5
6  typedef struct Project {
7      char* title;
8      char* description;
9      char* deadline;
10     //TaskList* task_list;
11     struct Project* next;
12 } Project;
13
14 typedef struct {
15     Project* head;
16 } ProjectList;
17
18 // Project management functions
19 Project* create_project(); // Takes user input inside
20 void add_project(); // Adds a new project from user input
21 void view_project();
22
23
24 #endif
```

4.task.h

Defines the structure for individual tasks, including attributes like title, description, and priority.

Sample Code(task.h):

```
1  #ifndef TASK_H
2  #define TASK_H
3  #include "priority.h"
4
5  typedef enum{
6      TODO,
7      IN_PROGRESS,
8      DONE
9  } TaskStatus;
10
11 typedef struct Task{
12     char* title;
13     char* description;
14     char* deadline;
15     Priority priority_level;
16     TaskStatus status;
17     struct Task* next;
18 } Task;
19
20 typedef struct {
21     Task* head;
22 } TaskList;
23
24 Task* create_task(const char* title, const char* description, const char* deadline, enum Priority priority_level);
25 void add_task(TaskList* list, Task* task);
26 void view_tasks(TaskList* list);
27 void edit_task(Task* task);
28 void delete_task(TaskList* list, const char* title);
29 void reorder_tasks(TaskList* list); // Optional for drag-drop behavior
30 void free_task(Task* task);
31 void free_task_list(TaskList* list);
32
33
34 #endif // TASK
```

5. priority.c / priority.h

Manages conversion between string and enum types for priorities and provides utilities for assigning priority levels.

Sample Code(priority.h):

```
1  #ifndef PRIORITY_H
2  #define PRIORITY_H
3
4
5  typedef enum {
6      PRIORITY_LOW = 1,
7      PRIORITY_MEDIUM = 2,
8      PRIORITY_HIGH = 3
9  } Priority;
10
11
12 const char* priority_to_string(Priority p);
13
14 #endif
```

6. utils.c / utils.h

Provides helper functions for common operations like string handling, screen clearing, and user input validation.

Sample Code(utils.h):

```
1  #ifndef UTILS_H
2  #define UTILS_H
3  #include <stdint.h>
4
5  char* get_string_input(const char* prompt);
6  uint16_t get_integer_input(const char* prompt);
7
8  #endif // UTILS_H
```

Data Structures Used

- **Linked Lists:**

- Projects and tasks are stored using singly linked lists to allow dynamic addition and removal without fixed size limitations.

- **Enums:**

- Task priorities are managed using an enumerated type (LOW, MEDIUM, HIGH), ensuring type safety and clarity.

Examples:

- a. **Enum:**

```
1  #ifndef PRIORITY_H
2  #define PRIORITY_H
3
4
5  typedef enum {
6      PRIORITY_LOW = 1,
7      PRIORITY_MEDIUM = 2,
8      PRIORITY_HIGH = 3
9  } Priority;
10
11
12  const char* priority_to_string(Priority p);
13
14  #endif
```

b. Linked List:

```
c.     if (!fgets(line, sizeof(line), file)){
        |     free(title);
        |     free(description);
        |     break;
    }
    line[strcspn(line, "\n")] = 0;
    char* deadline = malloc(strlen(line) + 1);
    strcpy(deadline, line);

    fgets(line, sizeof(line), file); // skip '---'

    Project* p = malloc(sizeof(Project));
    p->title = title;
    p->description = description;
    p->deadline = deadline;
    //p->task_list = NULL;
    p->next = NULL;

    projects[count++] = p;
}
```

File Handling

- A file named **projectlist.txt** stores the list of all projects and their corresponding tasks.
- Each time a change is made to projects or tasks, the data is saved to ensure persistence.
- Upon startup, the program loads this data to resume from where it was left off.

Example:

```
Web App
Build dashboard for admin users
2025-07-01
---
School Work
Finish OS project and revision
2025-06-15
---
Freelance Project
Client website design
2025-06-30
---s
```